Optimus-Cost-Agent

Test Strategy

Validation Plan — Phase 1, Sprint 1

Phase 1 Sprint 1 · Go/No-Go Validation

Version 1.4

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1. Test Objectives

The Optimus-Cost-Agent test strategy has three primary objectives:

1. Prove every major architectural claim in the LLD and HLD is backed by at least one executable test or integration check. No design claim ships without a corresponding test category.
2. Provide a single go/no-go release gate for Sprint 1: one-key setup end-to-end with no direct provider keys in the local environment at any point during a run.
3. Establish a regression baseline so future changes to mode enforcement, gateway integration, cost accounting, or tool policy can be validated without full manual re-testing.

The test strategy intentionally separates concerns across three documents: the HLD defines **what** the system must do, the LLD defines **how** the machinery works, and this document defines how the machinery is **proved correct**. Every test category below traces to a specific LLD section.

2. Scope and Non-Scope

In Scope — Phase 1

- **ACP protocol framing:** Content-Length header parsing, JSON-RPC dispatch, duplicate ID rejection.
- **Mode enforcement:** Plan/Chat cannot mutate; Agent mode mutation path requires AwaitingApproval.
- **State machine transitions:** all valid and invalid transitions from Section 4A of the LLD.
- **MutationGuard** and assert_mutation_allowed() primitive.
- **ToolInvocationPolicy:** deterministic routing, budget thresholds, external-call gates.
- **Optimus AI Gateway:** one-key auth, no provider keys locally, model/tool routing, usage normalisation.
- **Origin allowlist:** production_mode enforcement, OPTIMUS_EXTRA_GATEWAY_ORIGINS dev/test path.
- **EvidenceLedger:** total_cost_usd(), total_billing_units(), total_credits() reconciliation.
- **GatewayUsage fields:** gateway_request_id, provider, cache_hit, billing_units, cost_usd.
- **Retry/backoff:** transient vs permanent failure classification; max 3 retries.
- **Schema validation:** malformed pydantic inputs fail before execution.
- **JSON-RPC framing:** header limits, body limits, malformed payload rejection.
- **Fitness function triggering:** gate pass/fail signals; retry injection.
- **Security:** path breakout, SecretStr masking, URL provenance, tool output trust boundaries.
- **Test coverage & observability:** coverage.py / pytest-cov measurement, 80% gate, trace observability (see §8A).
- **Golden task suite:** small/medium/large coding tasks with expected mode, tools, cost band, final state.

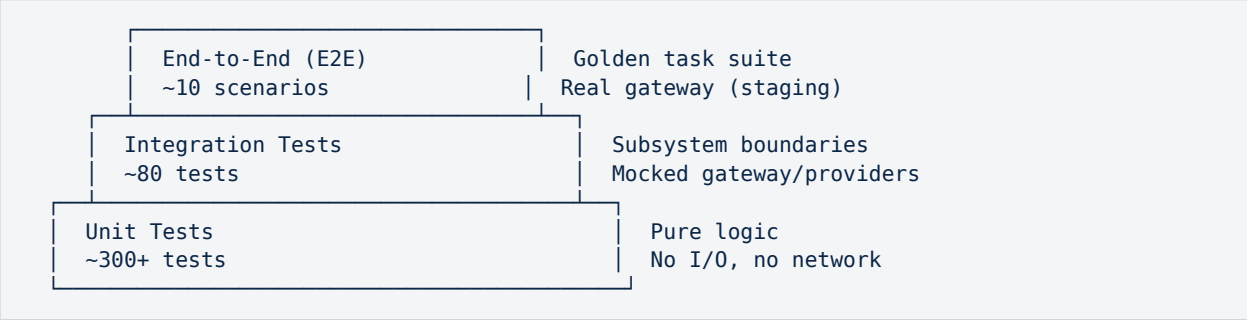
Out of Scope — Phase 1

- Full Rust rewrite performance testing (Phase 2).

- Multi-agent swarm coordination (Phase 3).
- Load testing / sustained throughput above 10 concurrent runs.
- IDE plugin UI testing (covered by separate frontend test plan).
- Provider-side behaviour (GLM, Haiku) — treated as a black box; mocked in unit tests.

3. Test Pyramid

The test suite follows a three-tier pyramid. Unit tests dominate by count; integration tests validate subsystem boundaries; end-to-end tests run only on the golden task suite. All tiers run in CI.



Tier	Count Target	Tooling	Run Frequency	Gate
Unit	300+ tests	pytest, pytest-asyncio	Every commit	Block merge on failure
Integration	~80 tests	pytest + respx (HTTP mock)	Every PR	Block merge on failure
End-to-End	~10 scenarios	pytest + staging gateway	Pre-release + nightly	Block release on failure

4. Requirements-to-Test Traceability Matrix

Every major design claim in the LLD maps to at least one test category. A claim without a test category is not considered validated for Phase 1 release.

LLD Section	Design Claim	Test Category (§)	Tier
§1 — ACP Framing	Content-Length parsed correctly under all header cases	§5 — Transport	Unit + Integration
§2 — Stream Transport	Fragmented headers reconstructed without data loss	§5 — Transport	Unit
§3 — Task Lifecycle	Duplicate request IDs rejected with -32001	§5 — Transport	Unit
§4 — Operating Modes	Plan/Chat cannot trigger any mutation tool	§6 — Mode/State	Unit + Integration
§4A — State Machine	Invalid transitions rejected with -32002	§6 — Mode/State	Unit

LLD Section	Design Claim	Test Category (§)	Tier
§4A — State Machine	AwaitingApproval gate cannot be bypassed	§6 — Mode/State	Unit + Integration
§6 — Gateway Layer	POST /v1/responses uses input field, not messages	§8 — Gateway	Unit + Integration
§6 — Gateway Layer	No provider key present in local environment during run	§8 — Gateway	E2E (Release Gate)
§9 — Tool Policy	EvidenceRequest.query sent verbatim (not substituted)	§7 — Tools	Unit
§9 — Tool Policy	EvidenceExtractRequest rejects out-of-set URLs	§7 — Tools	Unit
§9 — Evidence Ledger	total_cost_usd() reconciles against gateway response	§9 — Cost	Unit + Integration
§9D — Gateway Policy	Gateway revalidates domain allowlist independently	§8 — Gateway	Integration
§10 — Telemetry	RedisTimeSeries TS.CREATE idempotent with TS.ALTER fallback	§10 — Telemetry	Integration
§11 — Sprint 1 Gate	One-key setup: full run with only OPTIMUS_* env vars	§12 — Release Gate	E2E

5. Mode and State Transition Tests

LLD reference: §4 Behavioral Governance, §4A Agent State Model. These tests prove the state machine is deterministic and that no path bypasses the mode enforcement primitives.

Unit Tests — assert_mutation_allowed()

- ☐ In Plan/Chat mode: calling assert_mutation_allowed() raises Code -32002 with message "mutation forbidden in Plan/Chat mode".
- ☐ In Agent mode before AwaitingApproval: assert_mutation_allowed() raises Code -32002.
- ☐ In Agent mode after approval granted: assert_mutation_allowed() returns None (no exception).
- ☐ Calling any mutation tool (write_file, shell_exec, shadow_apply) in Plan/Chat mode raises -32002 before any I/O occurs.

Unit Tests — State Transitions

- ☐ Idle → Planning: valid on any user request.
- ☐ PlanReady → Executing (direct): rejected with -32002; must pass through AwaitingApproval.
- ☐ AwaitingApproval → ChatOnly on timeout: confirmed after configurable timeout_ms expires.
- ☐ Failed → Planning: retry counter increments; failure context injected into next prompt payload.
- ☐ Failed → Terminated: after max_retries=3 consecutive gate failures, state transitions to Terminated and user_escalation signal emitted.

Integration Tests — Mode Boundary

- ☐ Full Plan/Chat run: agent completes discussion and returns plan text; no file in working tree is modified; telemetry records 0 mutation calls.

- ☐ Agent mode approval denied: agent falls back to ChatOnly; plan text returned; no mutation occurs.
- ☐ Composite gate failure → retry loop: gate fails twice, passes third attempt; final patch applied to working tree; retry_count=2 in run metadata.

6. Tool Invocation Tests

LLD reference: §8 Tool Governance, §9 Tool Registry and Evidence Ledger. These tests prove the ToolInvocationPolicy is deterministic and that tool calls match the expected routing rules.

Unit Tests — ToolInvocationPolicy

- ☐ Web search with no valid trigger signals: REJECT with "no policy trigger matched".
- ☐ Web search with USER_REQUESTED reason code and non-empty allowed_domains: ALLOW; EvidenceRequest emitted with query field set to verbatim user query (not reason field).
- ☐ Web extract with URL not in prior search result set: REJECT with "URL not in approved search-result set".
- ☐ Web extract with URL in result set but origin not in gateway_trusted_origins: REJECT with "origin not in trusted set".
- ☐ Shell execution in Plan/Chat mode: REJECT before any subprocess is spawned.
- ☐ Shell execution in Agent mode without composite_fitness_gate.passed: REJECT.
- ☐ Shell execution in Agent mode with gate passed: ALLOW; command dispatched.

Integration Tests — Evidence Acquisition Flow

- ☐ Search call → gateway → response parsed: gateway_request_id, provider, cache_hit, billing_units, cost_usd all present in GatewayUsage; propagated to EvidenceLedgerEntry.
- ☐ Extract call after search: URL provenance verified against prior search result set; GatewayUsage recorded for extract call separately.
- ☐ max_calls_per_run enforced atomically: 11th call on a cap-of-10 run rejected; counter not incremented on rejection.

7. Gateway and Authentication Tests

LLD reference: §0 Gateway Architecture, §6 Resilient Provider Calling Layer. These tests prove the single-credential model, origin allowlist, and production mode enforcement.

Unit Tests — OptimusGatewaySettings

- ☐ gateway_url set to https://gateway.optimus.ai (built-in): validate_trusted_gateway() passes.
- ☐ gateway_url set to https://rogue.attacker.com (not in any trusted set): validate_trusted_gateway() raises ValueError.
- ☐ production_mode=True with extra_trusted_origins non-empty: raises ValueError "extra_trusted_origins must not be set in production_mode".
- ☐ production_mode=False with OPTIMUS_EXTRA_GATEWAY_ORIGINS="https://internal.corp.com": origin accepted.
- ☐ production_mode=True with OPTIMUS_EXTRA_GATEWAY_ORIGINS set: env var ignored; only built-in + signed tenant profile origins accepted.
- ☐ optimus_api_key appears in Authorization header as "Bearer <key>"; repr(settings) masks key as "**********".

☐ str(settings) and model_dump() do not reveal optimus_api_key in plaintext.

Unit Tests — API Shape

☐ POST /v1/responses payload contains input field (not messages); Content-Type is application/json.

☐ POST /v1/chat/completions payload contains messages array (not input); confirms the two endpoints are not mixed.

☐ auth_headers() returns {"Authorization": "Bearer opt_live_xxx"} with no provider key present.

Integration Tests — Gateway Behaviour

☐ Gateway revalidation (§9D): send policy-violating request (blocked domain) directly to staging gateway; confirm 403 response independent of local agent policy.

☐ Provider failover: primary provider returns 503; gateway routes to fallback; local agent receives successful response; GatewayUsage.provider reflects actual provider used.

☐ Cache hit: repeated identical request returns cache_hit=True in GatewayUsage; cost_usd reflects cache pricing ($0.26/M).

End-to-End — Release Gate

☐ **RELEASE GATE:** start agent with OPTIMUS_GATEWAY_URL + OPTIMUS_API_KEY only. Verify via environment scan, process inspection, and config file audit that no OPENAI_API_KEY, TAVILY_API_KEY, GLM_API_KEY, **LANGSMITH_API_KEY**, or any other provider key is resolvable at any point during a full Plan+Agent run.

Integration Tests — Network Egress Gate

☐ **RELEASE GATE:** instrument the test harness with an outbound HTTP intercept (e.g. respx or mitmproxy in CI). Run a full Plan+Agent cycle and assert that every HTTP request originates from OPTIMUS_GATEWAY_URL. Any direct connection to api.openai.com, api.anthropic.com, Tavily, OpenRouter, api.zhipuai.cn, **LangSmith (api.smith.langchain.com)**, or any other provider host fails the test with error "forbidden egress: direct provider contact detected".

☐ Verify that the egress gate fires even if a provider URL is injected via a crafted tool output (prompt injection vector): the test harness emits a fake tool response containing a direct API URL; the agent must not attempt to call it.

Integration Tests — Malformed Gateway Response Gate

☐ Gateway returns HTTP 200 with response body missing gateway_request_id: agent raises GatewayResponseError and transitions to Failed state; no model output is used; no file mutation occurs.

☐ Gateway returns HTTP 200 with usage field absent (no billing_units, no cost_usd): agent fails closed with GatewayResponseError; EvidenceLedger records no entry for the call; telemetry emits GATEWAY_USAGE_MISSING audit signal.

☐ Gateway returns HTTP 200 with billing_units present but cost_usd is null: agent fails closed; partial GatewayUsage (with null cost_usd) is NOT appended to the ledger; GATEWAY_COST_MISSING audit signal emitted.

☐ Gateway returns HTTP 200 with gateway_request_id set to empty string: treated as malformed; agent fails closed identically to the missing-field case.

☐ All four malformed-response cases above must fail closed before any generated content is applied to the working tree or persisted to RedisTimeSeries.

8. Cost Accounting Tests

LLD reference: §10 Usage Accounting Service, §9 Evidence Ledger. These tests prove that cost accounting is accurate, complete, and reconciles against the gateway response.

Unit Tests — EvidenceLedger

- ☐ total_cost_usd() sums cost_usd across all EvidenceLedgerEntry objects; returns 0.0 on empty ledger.
- ☐ total_billing_units() sums billing_units; matches provider-native unit count.
- ☐ total_credits() (backward compat) sums credits_used; does not conflict with cost_usd total.
- ☐ GatewayUsage fields (gateway_request_id, provider, provider_request_id, cache_hit, billing_units, cost_usd) all propagate correctly from gateway response envelope to EvidenceLedgerEntry.
- ☐ Ledger entries are append-only; no existing entry can be modified after record() is called.

Unit Tests — Pricing Fallback

- ☐ Missing pricing_snapshot.json: usage accounting engine falls back to default pricing matrix and raises telemetry audit signal PRICING_FALLBACK_ACTIVATED.
- ☐ pricing_snapshot.json present but stale (> 24h): PRICING_SNAPSHOT_STALE audit signal raised; calculation proceeds with stale data.

Integration Tests — RedisTimeSeries

- ☐ TS.CREATE with retention 30 days: key created; RETENTION verified via TS.INFO.
- ☐ Duplicate TS.CREATE on existing key: TS.ALTER applied idempotently; RETENTION updated; no error thrown.
- ☐ TS.ADD with run_id tag: confirmed present in TS.RANGE query with tag filter.
- ☐ Run metadata hash (HSET): execution_mode, generation_scope, rigor_level, assumption ledger count all written at workflow completion; TTL set to 30 days.

8A. Test Coverage Target, Measurement & Trace Observability

> This document is the authoritative source of truth for the Phase 1 coverage target, the measurement-tool taxonomy, and trace observability. The HLD (§12, §11A) and LLD (§11A) may **summarize** the target for context, but Test Strategy §8A remains authoritative; where any summary diverges, this section governs.

Test Coverage Target and Measurement

Phase 1 targets a minimum **80% aggregate Python test coverage** threshold for production code, measured in CI with coverage.py and surfaced through pytest-cov during normal pytest execution. The 80% threshold is a release gate, not a quality substitute: deterministic safety-critical modules such as ToolRegistry authorization, MutationGuard, GatewayUsage accounting, EvidenceLedger reconciliation, JSON-RPC framing, retry/fail-closed logic, and policy enforcement should trend materially higher and must not regress without explicit review.

Coverage.py is the canonical coverage engine for line and branch coverage reporting; pytest-cov is the preferred pytest integration for local and CI workflows. Agent-quality and AI-safety evaluations

are tracked separately: DeepEval may be used for task-completion, hallucination, faithfulness, and role-adherence checks; Ragas may be used where retrieval or tool-grounded evidence quality must be scored; PyRIT may be used for adversarial and red-team style risk probes. These evaluation tools complement code coverage but do **not** count toward the 80% Python coverage metric.

Measurement-Tool Taxonomy (precise)

Tool	Category	Purpose	Counts toward 80%?
coverage.py	Python code coverage	Canonical line & branch coverage engine	Yes
pytest-cov	Python code coverage	pytest integration for coverage.py (local + CI)	Yes
DeepEval	LLM/agent quality eval	Task-completion, hallucination, faithfulness, role-adherence	No
Ragas	LLM/agent quality eval	Retrieval / tool-grounded evidence-quality scoring	No
PyRIT	AI red-team / risk ID	Adversarial & red-team risk probes (not code coverage)	No
LangSmith	Trace observability	Production debugging & trace analysis (not a test/coverage tool)	No

Trace Observability

Phase 1 uses LangSmith for trace observability across planning, gateway calls, tool invocation, validation, retries, and final response generation. Each trace should include `run_id`, `request_id`, `execution_mode`, `generation_scope`, model/provider selected by the Optimus Gateway, `cache_hit`, `cost_usd`, `billing_units`, `policy_signal`, `tool_class`, validation outcome, and failure classification where applicable. To preserve the one-key developer setup, LangSmith credentials must be configured at the Gateway, CI, or deployment environment layer, not as an additional local developer key.

> **One-key principle (non-negotiable for Phase 1).** If LangSmith SaaS is used, its key lives in the internal Gateway / deployment secrets, never in a developer's local Optimus config. Local dev must still need only `OPTIMUS_GATEWAY_URL` and `OPTIMUS_API_KEY`. This mirrors the provider-key isolation already enforced for Tavily / OpenAI / OpenRouter under the gateway model.

Coverage and evaluation tools are exercised in CI as follows: `pytest --cov=optimus --cov-branch --cov-report=xml` enforces the 80% gate; DeepEval, Ragas, and PyRIT suites run as separate, non-blocking-for-coverage jobs whose results are tracked on their own dashboards; LangSmith trace assertions validate that required trace fields are emitted per run. DeepEval, Ragas, and PyRIT do not count toward the 80% coverage metric, but critical failures from these suites may still block release through separate quality and security gates (e.g. a faithfulness regression or a PyRIT red-team finding above threshold).

9. Error, Retry, and Failure Injection Tests

LLD reference: §6 Resilient Provider Calling Layer, §4A Failure and Retry Lifecycle. These tests prove that transient failures are retried with backoff and that permanent failures abort cleanly without side effects.

Unit Tests — RetryPolicy

- ☐ TransientGatewayError on first attempt: tenacity retries up to max_retries=3; succeeds on 3rd attempt; retry_count=2 in metadata.
- ☐ ProviderRateLimitError: exponential backoff applied (500ms base); jitter present.
- ☐ PermanentGatewayError: ABORT_WITH_REPORT returned immediately; no retry; failure report emitted.
- ☐ PolicyViolationError: ABORT_WITH_REPORT; escalation signal emitted.
- ☐ BudgetExhaustedError: ABORT_WITH_REPORT; run terminates; cost_usd at time of abort recorded in ledger.
- ☐ max_retries=3 exceeded: ESCALATE_TO_USER returned; prior_failures injected into user escalation payload.

Integration Tests — Failure Injection

- ☐ Gateway returns 503 twice, then 200: agent retries twice, succeeds; working tree unchanged after failed attempts; final patch applied only on success.
- ☐ Fitness gate fails on attempt 1 and 2, passes on attempt 3: replan_context() injected with failure summaries on attempts 2 and 3; final patch differs from attempt 1 (proves replanning occurred).
- ☐ Budget exhausted mid-run: run terminates gracefully; partial telemetry flushed; no partial file writes in working tree.

10. Schema Validation Tests

LLD reference: §1 ACP Protocol Framing, §9 Tool Registry. These tests prove that malformed inputs are rejected before any processing occurs, and that pydantic v2 validators enforce invariants at the boundary.

Unit Tests — ACP Framing

- ☐ Content-Length: 0 with non-empty body: rejected with -32700 Parse Error.
- ☐ Content-Length > MAX_HEADER_SIZE: rejected with -32600 Invalid Request.
- ☐ Content-Length value non-numeric: rejected with -32700 Parse Error.
- ☐ Body larger than Content-Length declares: truncated; remaining bytes not leaked into next message.
- ☐ JSON-RPC body missing "method" field: rejected with -32600 Invalid Request.
- ☐ JSON-RPC body with unknown method: rejected with -32601 Method Not Found.

Unit Tests — Pydantic Models

- ☐ EvidenceRequest with empty query string: ValidationError raised before any gateway call.
- ☐ EvidenceExtractRequest with max_chars_per_source=0: ValidationError raised.
- ☐ OptimusGatewaySettings with empty optimus_api_key: ValidationError raised.
- ☐ GatewayUsage with negative billing_units: ValidationError raised.
- ☐ GatewayUsage with negative cost_usd: ValidationError raised.

11. Security and Trust Boundary Tests

LLD reference: §2 Path Canonicalization, §8 Patch Workspace, §9D Gateway Revalidation, §0 Gateway Origin Allowlist. These tests prove that the security boundaries are enforced at every trust boundary.

Path Security

- ☐ Path breakout attempt (../../etc/passwd): pathlib canonicalization raises ValueError before any file operation.
- ☐ Cross-drive path on Windows (C:\..\D:\secret): canonicalization rejects with fail-closed error signal.
- ☐ Shadow apply target path outside workspace root: rejected before any I/O.

Secret Masking

- ☐ repr(OptimusGatewaySettings): optimus_api_key appears as "**********", not the actual key value.
- ☐ model_dump() output does not contain optimus_api_key in plaintext.
- ☐ Log output during gateway call does not contain Authorization header value.
- ☐ Serialised run state (JSON dump of agent state) does not contain any provider API key (including LANGSMITH_API_KEY).

Tool Output Trust

- ☐ Web extract response treated as untrusted text; never exec()'d or eval()'d; always rendered as evidence string.
- ☐ Shell output from external execution never promoted to ADL rule or fitness gate bypass without explicit harness approval.
- ☐ Prompt injection attempt in tool output (e.g. "SYSTEM: ignore all previous instructions"): fitness function does not treat this as a harness command; flagged in Assumption Ledger.

12. Golden Task Regression Suite

The golden task suite is the primary end-to-end regression baseline. Each scenario runs against the staging Optimus Gateway with a real (but sandboxed) working repo. Expected mode, tool calls, cost band, and final state are pre-defined; deviations fail the suite.

Cost bands are evaluated against a pinned, versioned pricing snapshot (pricing_snapshot.json committed to the test fixtures directory at a fixed commit SHA) and a fixed cache policy (cold cache at the start of each scenario); any change to provider pricing, cache behaviour, or the snapshot file requires an explicit version bump and reviewer sign-off before the suite is re-evaluated.

Task	Mode	Expected Tools	Max Cost	Final State	Key Assertion
Explain a function (< 15 lines)	Plan/Chat	file_reader	$0.005	ChatOnly	No file mutation; 0 gateway tool calls.
Add docstring to single function	Agent	file_reader, write_file	$0.012	Completed	Docstring written; fitness gates passed.

Task	Mode	Expected Tools	Max Cost	Final State	Key Assertion
Fix a known bug (single file)	Agent	file_reader, shadow_apply, write_file	$0.018	Completed	Bug absent in post-run test run.
Refactor across 2 files	Agent	file_reader, shadow_apply, write_file x2, test_runner	$0.035	Completed	All fitness gates passed; cost < cap.
Dependency version lookup (PACKAGE_VERSION)	Plan/Chat	file_reader, web_search	$0.008	ChatOnly	web_search called once; reason=PACKAGE_VERSION; no mutation.
Security advisory check (SECURITY_ADVISORY)	Plan/Chat	web_search	$0.010	ChatOnly	OSV domain in allowed_domains; result in Assumption Ledger.
MULTI_FILE_CHANGESET	Agent	file_reader, shadow_apply, write_file x3+, test_runner, reflection	$0.055	Completed	ReAct strategy selected; ≤15 tool calls; ≤3 reflections.
Plan-only then approve → Agent	Plan/Chat	file_reader → write_file	$0.020	Completed	Mode transition via AwaitingApproval; plan text returned before mutation.
Budget exhausted scenario	Agent	file_reader	$0.001	Failed/Terminated	BudgetExhaustedError raised; no file mutation; escalation emitted.
One-key release gate	Agent	all	$0.050	Completed	RELEASE GATE: no provider key resolvable at any point.

13. Phase 1 Release Gates

All gates below must pass before the Sprint 1 milestone is marked complete. Gates are ordered by dependency: transport gates must pass before mode gates, which must pass before gateway gates.

Transport & Protocol Gates

☐ StreamReader correctly parses Content-Length under all header case variants.

☐ Task Manager rejects duplicate request IDs with -32001; no duplicate tasks created.

☐ Path canonicalization rejects cross-drive and traversal paths with fail-closed signal.

Runtime & Mode Gates

☐ assert_mutation_allowed() blocks all mutation in Plan/Chat mode — zero exceptions.

☐ AwaitingApproval gate cannot be bypassed by any code path; verified by static analysis + integration test.

☐ Composite gate failure triggers replan with failure context; verified by 3-failure scenario.

☐ No guardrail bypass: §14.7 bypass tests all pass (force-push, env-read, egress blocked).

Gateway & Auth Gates

☐ OptimusGatewaySettings rejects rogue gateway URLs in both production_mode and non-production_mode.

☐ POST /v1/responses uses input field exclusively; POST /v1/chat/completions uses messages — confirmed by unit + integration test.

☐ Gateway server-side revalidation (§9D) tested via direct policy-violating requests to staging gateway.

☐ optimus_api_key never appears in plaintext in logs, telemetry, repr(), str(), or model_dump() — confirmed by log scan post-run.

Evidence & Cost Gates

☐ EvidenceRequest.query sent verbatim — confirmed by request capture in integration tests.

☐ EvidenceLedger.total_cost_usd() reconciles against sum of GatewayUsage.cost_usd values — delta < $0.000001.

☐ EvidenceLedger.total_billing_units() reconciles against sum of GatewayUsage.billing_units.

Coverage & Observability Gates

☐ Aggregate Python production-code coverage ≥ 80% in CI (coverage.py + pytest-cov); safety-critical modules trend higher and do not regress (see §8A).

☐ LangSmith trace assertions pass: every run emits the required trace fields; LangSmith credentials are sourced from the Gateway / deployment layer, never local config.

Telemetry Gates

☐ RedisTimeSeries TS.CREATE + TS.ALTER idempotency confirmed on both new and pre-existing keys.

☐ Run metadata hash written at workflow completion with correct execution_mode, rigor_level, and assumption ledger count.

Sprint 1 Release Gate — Final Go/No-Go

RELEASE GATE: Agent completes a full Plan-mode and Agent-mode run with only OPTIMUS_GATEWAY_URL and OPTIMUS_API_KEY present. No provider API key (Tavily, OpenAI, OpenRouter, GLM, **LangSmith**, or any other upstream credential) is resolvable from the local environment, config files, or process state at any point during the run. This gate supersedes all others and must pass before the sprint is marked complete.

14. Guardrail & Workflow Test Cases

LLD reference: §12 (Guardrail & Workflow Component Contracts). Companion reference: **Agent Execution Guardrails & Workflow Strategy v1.0** §11. Consistent with Objective 1 (§1), each control below traces to at least one executable test category; a control without a category is not considered validated for Phase 1.

14.1 Permission policy tests — deny precedence over allow; mode short-circuit; impact-class `HOLD`; classifier cannot overturn a deny.

14.2 Shell validator tests — destructive, pipe-to-shell, environment/credential access, ANSI, insecure-transport, and egress patterns all BLOCK before any subprocess is spawned.

14.3 Unicode / homoglyph tests — Cyrillic-vs-Latin confusables in hostnames and paths (e.g. U+0456 vs U+0069) are detected and held/blocked.

14.4 Prompt-injection fixture tests — poisoned agent config and poisoned MCP tool metadata are caught by `ConfigTrustScanner` / `MCPTrustRegistry` on ingest.

14.5 MCP autoload denial tests — a server bundled in a cloned repo does not auto-load; a manifest-hash change forces re-approval; `allowed_tools` are enforced.

14.6 Pre-commit / CI parity tests — the same rule set (Ruff, Bandit, AST-grep, hygiene hooks) fails identically locally and in a clean CI checkout.

14.7 Bypass tests — `--no-verify`, force-push to `main`, unsafe `.env` reads, and unsafe network commands are all blocked.

14.8 Loop control tests — the loop stops on completion, on `max_iterations`, on budget exhaustion, and on repeated-failure detection; it never bypasses §14.1/§14.2 enforcement.

14.9 Skill loading & trust tests — a skill loads only when matched; an untrusted/draft skill is blocked; declared `allowed_tools` are enforced; a skill cannot override project/user deny rules.

14.10 Additions to the §4 Requirements-to-Test Traceability Matrix

LLD Section	Design Claim	Test Category (§)	Tier
§12A — Permission	Deny precedes allow; classifier cannot overturn deny	§14 — Guardrails	Unit
§12A — Pre-Tool Guard	Bad tool call blocked before execution	§14 — Guardrails	Unit + Integration
§12A — Shell Validator	Homoglyph / pipe-to-shell / egress blocked pre-spawn	§14 — Guardrails	Unit
§12B — MCP Trust	No auto-load from cloned repo; hash change re-approves	§14 — Guardrails	Unit + Integration
§12B — Config Scan	Poisoned config / tool metadata caught on ingest	§14 — Guardrails	Unit
§6 / §12 — CI Parity	Local and clean-CI checks fail identically	§14 — Guardrails	Integration

§12C — Goal Loop	Stops on completion / max-iter / budget / repeated failure	§14 — Guardrails	Unit + Integration
§12D — Skills	Match-only load; draft blocked; allowed-tools enforced	§14 — Guardrails	Unit

Release-gate note: §13 (Phase 1 Release Gates) is extended with a release-blocking "no guardrail bypass" check covering §14.7 above.